

10(a)(i)	cannot predict when a (particular) nucleus will decay or cannot predict which nucleus will decay next	B1
10(a)(ii)	not affected by external / environmental factors	B1
10(b)(i)	line fluctuates or trend is a straight line	B1
10(b)(ii)	straight line of best fit drawn on Fig. 10.1	B1
10(b)(iii)	$M = M_0 \exp(-\lambda t)$ so $\ln M = \ln M_0 - \lambda t$ so gradient = $-\lambda$ (and magnitude of gradient = λ)	B1
10(b)(iv)	gradient = $(-)(8.0 - 4.8) / (11.6 - 0)$ (allow any correct pair of values from Fig. 10.1)	C1
	$\lambda = 0.28 \text{ s}^{-1}$	A1
10(b)(v)	half-life = $0.693 / \lambda$ = $0.693 / 0.28$ = 2.5 s	A1
10(c)	(for reaction to occur,) energy is released	B1
	energy release comes from fall in mass so total mass of products must be less (than mass of carbon-15)	B1